

# Comparing Distributions and Compound Events

## Answer Key

### Grade 6–7 Statistics and Probability

#### Quick Answers

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- |         |                   |                       |
|---------|-------------------|-----------------------|
| 1. 15.5 | 5. $\frac{1}{4}$  | 9. $\frac{5}{33}$     |
| 2. 15.5 | 6. 9              | 10. 40, $\frac{3}{8}$ |
| 3. 9    | 7. $\frac{1}{16}$ | 11. $\frac{39}{64}$   |
| 4. 19   | 8. 17             | 12. 4, $\frac{3}{32}$ |
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#### Curriculum

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**Level:** Grade 6–7 Statistics and Probability

6.SP.A – *problem 12*

6.SP.B.5 – *problems 1, 2, 3, 4, 6, 8, 12*

7.SP.C.7 – *problems 5, 10*

7.SP.C.8 – *problems 7, 9, 11, 12*

*Difficulty 1–5 of 5 across 15 tagged checks.*

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#### Common wrong answers

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2. *If they got 15: read the middle of the printed list as a single value instead of averaging the two middle values*
6. *If they got 14: used the greatest value instead of the upper quartile, so the answer became a partial range rather than an interquartile range*
8. *If they got 19.4286: divided the correct total by 7 instead of by the 8 scores in the set*
11. *If they got 0.75: added the two single-spin probabilities instead of subtracting the no-red case from 1*
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#### 1. Median of Team Alpha.

The list is already in order: 12, 14, 15, 15, 16, 18, 19, 21. With an even count (8 values) the median is the mean of the two middle values, the 4th and the 5th.

$$\frac{15 + 16}{2} = \frac{31}{2} = 15.5$$

|      |
|------|
| 15.5 |
|------|

**2. Median of Team Bravo.**

In order: 8, 11, 15, 15, 16, 20, 24, 27. The 4th and 5th values are again 15 and 16.

$$\frac{15 + 16}{2} = 15.5$$

|      |
|------|
| 15.5 |
|------|

*Common wrong answer 15:* one middle value was reported on its own. An even-sized set has *two* middle values and the median sits between them — here the median is not even a score anyone earned.

*Worth noticing:* both teams have exactly the same median. Center alone cannot tell these two distributions apart — that is what problem 12 is about.

**3. Range of Team Alpha.**

Range is the greatest value minus the least value.

$$21 - 12 = 9$$

|   |
|---|
| 9 |
|---|

**4. Range of Team Bravo.**

$$27 - 8 = 19$$

|    |
|----|
| 19 |
|----|

Bravo's range is more than twice Alpha's, even though the two medians match.

**5. One marble, probability of blue.**

A single draw is a simple probability: favourable outcomes over total outcomes. The bag holds  $5 + 3 + 4 = 12$  marbles, and 3 of them are blue.

$$P(\text{blue}) = \frac{3}{12} = \frac{1}{4}$$

|               |
|---------------|
| $\frac{1}{4}$ |
|---------------|

**6. Interquartile range of Team Bravo.**

Split the ordered list at the median into two halves of four values each:

$$\underbrace{8, 11, 15, 15}_{\text{lower half}} \quad \bigg| \quad \underbrace{16, 20, 24, 27}_{\text{upper half}}$$

$Q_1$  is the median of the lower half:  $\frac{11 + 15}{2} = 13$ .

$Q_3$  is the median of the upper half:  $\frac{20 + 24}{2} = 22$ .

$$\text{IQR} = Q_3 - Q_1 = 22 - 13 = 9$$

9

*Common wrong answer 14:* the greatest value 27 was used in place of  $Q_3$ , giving  $27 - 13$ . The IQR deliberately ignores the extremes — that is the whole reason it exists beside the range.

**7. Spinner red AND cube shows 5.**

Two stages happen together, so this is a compound event. The spin does not change the cube, so the stages are independent and the probabilities multiply.

Spinner: 3 of the 8 equal sections are red, so  $P(\text{red}) = \frac{3}{8}$ .

Cube: one face out of 6 shows a 5, so  $P(5) = \frac{1}{6}$ .

$$\frac{3}{8} \times \frac{1}{6} = \frac{3}{48} = \frac{1}{16}$$

$\frac{1}{16}$

*Grading note:* a student who adds instead of multiplying gets  $\frac{3}{8} + \frac{1}{6} = \frac{13}{24}$ , a probability far larger than either stage — a useful sanity check, since “both” can never be more likely than either part alone.

**8. Mean of Team Bravo.**

Add all eight scores, then divide by how many there are:

$$8 + 11 + 15 + 15 + 16 + 20 + 24 + 27 = 136, \quad \frac{136}{8} = 17$$

17

*Common wrong answer 19.43:* the total was divided by 7. Count the values after adding them — there are 8 scores, so the divisor is 8.

Bravo’s mean (17) sits above its median (15.5) because the two high scores, 24 and 27, pull the mean upward. The median does not move.

**9. Two red marbles, no replacement.**

Compound event again, but the first draw changes the second, so the second stage must be recalculated from what is left.

First draw: 5 red out of 12 marbles, so  $P = \frac{5}{12}$ .

Second draw: one red marble is gone, so 4 red remain out of 11 marbles, giving  $P = \frac{4}{11}$ .

$$\frac{5}{12} \times \frac{4}{11} = \frac{20}{132} = \frac{5}{33}$$

$$\boxed{\frac{5}{33}}$$

*Grading note:* using  $\frac{5}{12}$  twice gives  $\frac{25}{144}$ , which is the *with*-replacement answer. The phrase “not put back” is what changes the second fraction.

**10. Club sign-ups.**

(a) “Altogether” means every bar counts:

$$9 + 6 + 15 + 10 = 40$$

$$\boxed{40}$$

(b) Now the 40 slips are the whole sample space and the 15 Robotics slips are the favourable outcomes:

$$P(\text{Robotics}) = \frac{15}{40} = \frac{3}{8}$$

$$\boxed{P = \frac{3}{8}}$$

*Grading note:* the total has to be computed before the probability can be written — a student who uses 4 (the number of clubs) as the denominator has counted categories instead of students.

**11. At least one red in two spins.**

“At least one” covers three different outcomes (red then not-red, not-red then red, red then red), so counting them directly is slow and error-prone. Its opposite is a single case: *neither* spin is red.

Not red on one spin:  $8 - 3 = 5$  of the 8 sections, so  $P = \frac{5}{8}$ .

The two spins are independent (the spinner has no memory):

$$P(\text{no red at all}) = \frac{5}{8} \times \frac{5}{8} = \frac{25}{64}$$

$$P(\text{at least one red}) = 1 - \frac{25}{64} = \frac{39}{64}$$

$$\boxed{\frac{39}{64}}$$

*Common wrong answer  $\frac{3}{4}$ :* the two single-spin probabilities were added,  $\frac{3}{8} + \frac{3}{8}$ . That double-counts the outcome where both spins are red, and adding probabilities this way can even pass 1.

**12. Synthesis — comparing the two distributions.**

(a) Split Alpha's ordered list at its median:

$$\underbrace{12, 14, 15, 15}_{\text{lower half}} \quad | \quad \underbrace{16, 18, 19, 21}_{\text{upper half}}$$

$$Q_1 = \frac{14 + 15}{2} = 14.5, \quad Q_3 = \frac{18 + 19}{2} = 18.5.$$

$$\text{IQR} = 18.5 - 14.5 = 4$$

4

(b) Model answer: Team Alpha is more consistent. Both teams have the same median, 15.5, so the center says nothing about which team is steadier. The spread does: Alpha's IQR is 4 against Bravo's 9, and Alpha's range is 9 against Bravo's 19. Alpha's middle half is packed into a 4-point band and its whole set fits in 9 points, while Bravo's scores stretch from 8 to 27. On the dot plots, Alpha's dots cluster around the middle and Bravo's trail off in both directions.

(c) Two independent choices, one from each list, so multiply.

Alpha scores greater than 18: 19 and 21, so 2 of the 8 —  $P = \frac{2}{8}$ .

Bravo scores greater than 18: 20, 24 and 27, so 3 of the 8 —  $P = \frac{3}{8}$ .

$$\frac{2}{8} \times \frac{3}{8} = \frac{6}{64} = \frac{3}{32}$$

$$P = \frac{3}{32}$$

*Manual review:* part (b) is a written comparison, read by a teacher. Parts (a) and (c) are machine-checked. Credit (b) when the student says the medians are equal AND uses at least one spread measure to name Alpha as the more consistent team.